Seeing Through Deepfakes

A Detection System Using Transparency Signals in Generative AI

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Abstract

This report presents a work placement project with the Malta Digital Innovation Authority on AI-generated image and deepfake detection. A proof-of-concept was developed using transparency support tool combining metadata analysis, a fine-tuned visual classifier, decision fusion and conditional semantic analysis. It was constructed as an interpretable decision support pipeline, not a standalone forensic detector. The Evaluation revealed that metadata and fine-tuning were helpful on controlled data, but the fused system struggled with realistic deepfake-style samples when the visual classifier was confident and wrong. This highlights both the value and fragility of multi-signal transparency approaches, while providing a basis for further MSc research.

1. Introduction

Generative artificial intelligence (AI) is evolving rapidly, creating both opportunities and new emerging risks. These risks include misinformation, impersonation and manipulation of digital media. As AI-generated images and deepfakes become increasingly realistic, it is becoming harder for citizens, regulators and organisations to differentiate between authentic and manipulated content. In response, emerging European regulatory initiatives including the EU AI Act, transparency guidelines and code of practice are placing greater emphasis on clear labelling, provenance and accountability mechanisms for AI-generated content.

This work placement was carried out in collaboration with the Malta Digital Innovation Authority (MDIA). It focused on designing and developing a proof-of-concept AI-generated image detection and transparency support tool. The project explored combining multiple transparency signals like metadata provenance indicators and model-based visual classification to identify AI-generated or manipulated images. Rather than a single detection method, it investigates a modular and explainable approach where metadata flags provenance issues, visual classification identifies generative patterns, and fusion combines both into a weighted confidence assessment the user can follow.

1.1 Defining the Problem

The detection of AI-generated and manipulated image content is a technically challenging and societally relevant problem. Visual-only detection models will be unlikely to generalise across fast evolving generative models, and metadata and provenance signals can be absent, incomplete or intentionally removed. Since no single signal is reliable on its own, detection systems need to combine multiple sources of evidence and present their results transparently.

Therefore, the main issue during the placement was to study how a practical prototype could incorporate different detection components in a coherent pipeline. This included metadata and provenance analysis, visual classification, decision-level fusion and explainability support via visual attention outputs. The challenge was to make the system work technically, but also to make the output of the system understandable for any user.

1.2 Motivation

The motivation for this project arises from the growing societal impact of synthetic media. As AI-generated images become easier to produce and distribute, there is an increasing need for tools that can support transparency, trust and evidence-based decision making. This is especially true in regulatory contexts where public authorities and bodies of digital governance need to understand both the capacities and limitations of technical detection systems.

This role enabled an opportunity to link technical development with regulatory awareness. Engagement with EU-level discussions on transparency for AI-generated content helped frame the prototype as more than a purely technical classifier. Instead, the work aimed to explore how detection outputs could support broader transparency objectives by combining measurable signals and presenting them in an interpretable format.

1.3 Aim & Objectives

This work placement focused on the design, implementation and evaluation of a proof-of-concept system to detect AI-generated or manipulated images using a multi-signal approach. The prototype combined metadata, provenance information and visual classification into a modular pipeline, providing outputs that were designed to be interpretable and useful to regulators in their decision-making.

To do this, the placement worked through three connected objectives:

1. Analyse relevant regulatory and policy developments concerning transparency for AI-generated and manipulated content and assess how far existing frameworks address the risks involved.
2. Develop a modular prototype integrating metadata and provenance analysis, model-based visual classification and decision-level fusion.
3. Evaluate the prototype outputs with respect to detection performance, interpretability and practical limitations in a regulatory decision-support context.

2. Background

2.1 Synthetic Media, Trust & Regulatory Transparency

Synthetic media should be understood not merely as a technical artefact, but as a challenge to evidential trust. Generative AI systems have reduced the skill, time and cost required to produce realistic images, allowing persuasive visual content to be created and circulated at scale. This matters because images are still commonly treated as records of reality. While synthetic or manipulated images appear credible, they can instil uncertainty about authenticity, evidence and representation. Leibowicz therefore frames synthetic media as a governance problem, since it affects how audiences interpret online content and whether visual media can continue to function as a reliable marker of events, people or places [1]. Similarly, when images are reposted, compressed or detached from their original context, their provenance may become unclear, making it more difficult to assess whether an image is authentic, edited or AI-generated [2].

The European regulatory response increasingly treats the problem of verifying whether images are authentic or AI-generated as one of transparency rather than detection alone, and this project follows that framing directly by prototyping a disclosure interface that fills the gap between what Article 50 of the EU AI Act requires of providers and what audiences actually see, since disclosure mechanisms address how audiences are informed about synthetic content rather than requiring technical tools most users cannot access. Article 50 of the Act requires providers of AI systems that generate synthetic audio, image, video or text content to ensure, as far as technically feasible, that outputs are marked in a machine-readable format and detectable as artificially generated or manipulated. It also introduces disclosure expectations for deployers in circumstances where users may otherwise be misled into treating synthetic or manipulated content as authentic [3]. The Commission’s Draft Guidelines on Article 50 further indicate that these duties are intended to support clearer and more consistent compliance by providers, deployers and competent authorities [4]. Transparency is therefore not simply a user-interface feature, but part of a broader regulatory logic for accountability and supervision.

The definition of 'deepfake' in Article 3(60) of the EU AI Act is also relevant here, as it defines the problem more broadly than just manipulation of faces. It defines deepfakes as *"AI-generated or manipulated image, audio or video content that resembles existing persons, objects, places, entities or events, and could falsely appear authentic or truthful"* to a person [3]. This means that the risk of deepfakes cannot be reduced to a face-detection problem. It also has a semantic dimension, for an image may be misleading because of the person, place, building or event-like scene it appears to depict. For this reason, the conditional analysis should include more than just face-like content. However, this should be understood as only a partial response to the full scope of the definition, rather than a complete implementation of it.

Building on that regulatory logic, the Code of Practice on Marking and Labelling of AI-generated Content develops it in more operational terms by supporting a layered approach to transparency. Rather than relying on one signal, it refers to combinations of machine-readable marking, labelling, provenance information, watermarking and detection mechanisms [5]. This layered framing is important as individual signals are still imperfect: metadata can be missing or stripped, provenance can be partial, labels can be too coarse, and visual detectors can fail on unseen generators or degraded images [6]. A regulatory tool must therefore expose uncertainty and combine evidence, rather than present detection as a definitive judgement.

Within this placement, the collaboration with the MDIA gave the project a regulatory support purpose. This creates a need not only to understand legal requirements in abstract terms, but also to assess how they can be operationalised in practice. The prototype was therefore designed from the outset as a capability-building tool, not a production-ready enforcement system, and the evaluation reflects what such a tool can demonstrate rather than prove definitively. Instead, its value lies in demonstrating how transparency signals can be inspected, combined and presented in an interpretable manner. For MDIA, this supports practical understanding of the feasibility, limitations and supervisory relevance of marking, labelling, provenance and detection mechanisms, helping to inform how such tools might contribute to oversight, guidance and compliance monitoring as the EU AI transparency framework develops.

2.2 Technical Detection Approaches & Their Limitations

Technical approaches to AI-image detection are most commonly framed as supervised, real vs AI-generated classification tasks. In this setting, a model is trained on labelled examples of authentic and AI-generated images and learns to distinguish between them by identifying visual artefacts, statistical regularities or frequency-level inconsistencies associated with synthetic generation. Earlier approaches relied on manually defined forensic cues, but contemporary systems increasingly use deep learning models because they can learn subtle image-level patterns that may not be perceptible to human observers. Convolutional Neural Networks (CNNs), including architectures such as XCeption, ResNet and VGG16, remain central within the deepfake detection literature, particularly because they can capture spatial artefacts introduced by manipulation or generation processes [7]. More recent work also includes Vision Transformer based models, which is part of a broader trend towards architectures that can learn richer global representations of images rather than relying on local convolutional features alone [8].

The field is strongly benchmark driven and this influences the way detection systems are usually evaluated. Datasets such as GenImage, ArtiFact, WildFake and AI-GenBench have been developed to support systematic training and evaluation across real and AI-generated images, including images produced by different families of generative models [6][8][9][10]. These benchmarks are valuable because they allow models to be compared under controlled conditions and provide evidence that detectors can learn meaningful distinctions between authentic and synthetic imagery. However, benchmark performance must be interpreted cautiously, as results vary across datasets. High accuracy on a curated dataset does not necessarily imply reliable performance once a system is exposed to real world images that have been compressed, cropped, re-saved, reposted, edited or generated by newer models not represented in the training data.

The central technical limitation is generalisation, a problem multiple studies have approached differently. A detector may appear accurate because it has learned the artefacts of a specific generator, dataset or image collection, rather than a stable and transferable representation of synthetic origin. This is a fundamental challenge as highlighted by Rahman et al., Abbasi et al. and Pellegrini et al. who all point to the same issue: detectors tend to fail when tested out-of-distribution, be it unseen generators, different datasets, adversarial perturbations or newer generative models [6][7][8]. This is especially pertinent in a regulatory context, since synthetic media encountered by users, platforms or authorities will rarely come in the same controlled form as benchmark samples.

In addition to the distribution mismatch, visual classifiers are also plagued by the fact that their outputs are probabilistic, reflecting statistical likelihood rather than certainty about any given image. A confidence score indicates that an image resembles patterns associated with AI-generated or authentic images under the model's training assumptions. It does not prove who created the image, which tool was used, whether the image has been modified, or whether its provenance has been preserved. Tariq et al. thus criticises conventional deepfake detectors for acting as black-box systems that return binary labels or confidence scores with little transparency, which limits their usefulness in high-stakes contexts such as journalism, legal assessment and digital forensics [11]. Leibowicz adds that deploying detectors in isolation can be problematic because adversarial dynamics allow bad actors to adapt and evade technical interventions [1].

Visual detection remains useful but insufficient without supporting signals such as metadata and provenance. It provides an important content-based signal, especially where metadata or provenance information is absent, but it should not be treated as forensic proof or regulatory determination. In this placement, this limitation directly shaped the prototype design. The visual classifier was therefore implemented as one probabilistic evidence stream within a wider transparency pipeline, alongside metadata, provenance indicators, decision fusion and explainability support. This design reflects the view that detection should support human interpretation by exposing uncertainty and combining signals, rather than replacing judgement with a single automated verdict, which directly informs how the fusion, explainability, and provenance modules are structured in later sections.

2.3 Provenance, Fusion & Explainable Decision-support

Visual classifiers are more likely to infer synthetic origin from image content while provenance based approaches try to embed evidence into the media lifecycle itself. Metadata, provenance records, C2PA Content Credentials, watermarking and perceptual hashing thus provide complementary transparency signals rather than alternatives to visual detection. Metadata may reveal software traces, camera information, editing history or AI-related generation markers. Provenance mechanisms, by contrast, attempt to document the origin and subsequent transformation of a digital asset in a more structured manner. C2PA-style Content Credentials are especially relevant because they bind provenance assertions to media through cryptographically signed manifests, thereby supporting a more traceable account of who created or modified an image, when this occurred, and through which tools [2][12].

However, provenance should not be mistaken for truth itself. Content Credentials can demonstrate that a provenance record is well-formed, associated with an asset and free from tampering, but they do not automatically establish that every underlying claim is substantively true. A signed manifest is therefore better understood as authenticated evidence about a declared content history, rather than as a definitive guarantee of authenticity. This distinction is important in regulatory and forensic contexts, where a valid technical signal may still require interpretation. Nemecek et al. further show that provenance and watermarking layers can become desynchronised, producing cases where a cryptographically valid provenance record conflicts with pixel-level watermark evidence [13]. This reinforces the need to inspect signals together rather than trusting any single verification layer in isolation.

Watermarking and hashing address related, but narrower, transparency functions. Watermarking embeds perceptible or imperceptible signals into content so that a verifier can later identify a work as AI-generated or associate it with a particular provider. Perceptual hashing and fingerprinting instead reduce content to compact identifiers that can be matched against repositories of known material. These techniques are valuable within controlled ecosystems, and the Code of Practice on Marking and Labelling of AI-generated Content supports a layered approach involving machine-readable marking, metadata, provenance, watermarking and detection mechanisms [5]. Nevertheless, they remain incomplete. Metadata may be stripped during platform upload or re-sharing, provenance chains may be partial, watermarking can be degraded by compression, cropping or re-saving, and hashing depends on the availability of reference databases [2][14][15].

These limitations provide the rationale for decision fusion, much as prior multimodal work combined signals to offset gaps in any single channel, though here fusion is applied across metadata, provenance, watermark, and classifier evidence rather than audio-visual streams. Since no individual signal is sufficiently reliable across all contexts, a transparency-oriented detection system must combine imperfect evidence rather than privilege one source as authoritative. In multimedia forensics, fusion is commonly used to integrate evidence from different modalities or analytical components so that one signal may compensate when another is weak or unavailable. Salvi et al., for instance, demonstrate the value of combining audio and visual features for more robust deepfake detection [16]. Although this placement focuses on still images rather than multimodal media, the same principle applies, as metadata, provenance indicators, watermark traces and visual classifier outputs each capture different aspects of possible synthetic origin. Decision fusion should therefore be understood as an evidence aggregation method that surfaces uncertainty, not as a guarantee of correctness.

Explainability is the final requirement that makes such fusion usable. If a system takes multiple signals but only outputs a binary label, then it would reproduce the limitations of black box detection. Tariq et al. point out that deepfake detection tools for non-expert and high-stakes users should combine prediction and explanation, e.g. through confidence scores, visual saliency and narrative interpretation [11]. This aligns with the wider view that AI should support human judgement rather than replace it [17]. Accordingly, the prototype developed in this placement directly addresses the research question by treating detection as decision-support, aggregating evidence, preserving uncertainty and presenting the reasoning behind its assessment, while leaving final interpretation to the human user.

3. Methodology

3.1 System Architecture

The prototype was not built as a single binary classifier, but as a modular, multi-stage decision-support pipeline. As shown in Figure 1, the system begins with an uploaded image, which is analysed through two parallel components, namely a metadata and provenance module, and a visual classifier. This separation was necessary because each component captures a different type of transparency signal. This separation was necessary because each component captures a different type of transparency signal. The metadata module inspects embedded file information, software traces and provenance-related indicators, while the visual classifier estimates whether the image visually resembles real or AI-generated content. The outputs from both modules are then passed into a decision fusion stage, where the available evidence is combined into an overall confidence assessment. If the image is flagged as AI-generated or suspicious, the system moves to a conditional deepfake analysis module, which checks if the content may contain faces or places that look alike. The final result is presented through the user interface with a label, confidence score and explanatory information.

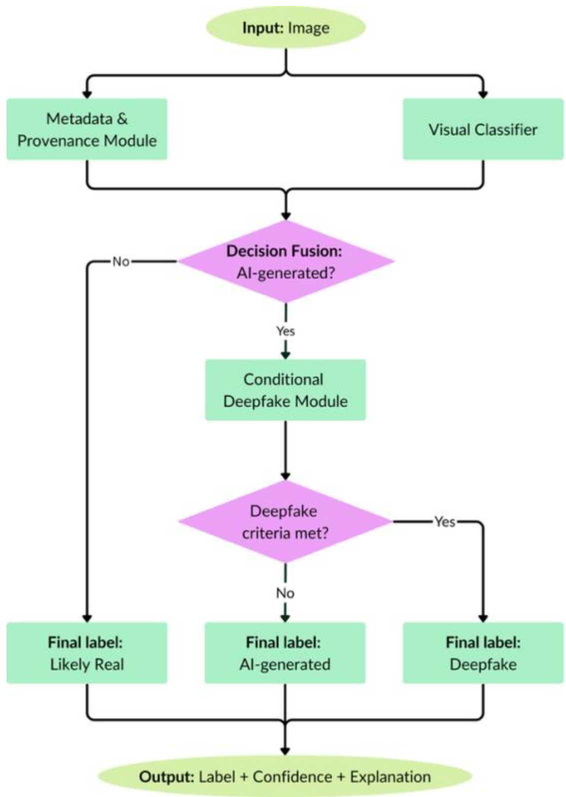

Figure 1: Multi-Staged Pipeline Architecture

This architecture reflects the layered transparency approach discussed in the Code of Practice [5]. Marking, labelling, provenance and detection mechanisms are treated as complementary rather than individually definitive. The prototype should therefore be understood as a transparent decision-support tool, not an autonomous forensic verdict.

3.2 Metadata & Visual Classification Modules

The metadata module was designed to inspect machine readable provenance and software traces that may indicate whether an uploaded image is AI-generated, edited or captured through a conventional device. It examines embedded EXIF fields, XMP namespaces, software tags, C2PA-style indicators and generator-related traces, before producing a metadata-derived AI-origin probability that is passed to the decision fusion stage. This evidence stream is useful because metadata can sometimes provide a more direct indication of origin than visual artefacts alone. For example, a file containing traces of a known generative tool or provenance related information may offer stronger contextual evidence than a visual prediction based only on image appearance. However, metadata was treated as fragile and incomplete rather than definitive. Online platforms may strip metadata during upload, compression or re-sharing, while users may also remove, overwrite or modify embedded information. Provenance records can also be partial or misleading, meaning that the absence of metadata does not prove that an image is authentic. For this reason, the metadata module was used as one structured evidence stream within the wider pipeline, rather than as an independent decision mechanism.

The visual classification module provides the second primary evidence stream by analysing the image content directly. It predicts if the uploaded image looks visually similar to real or AI-generated contents by a fine-tuned Vision Transformer classifier, reflecting the widespread use of transformer-based architectures for learning global image representations. For reproducibility, the classifier was based on the *'dima806/ai_vs_real_image_detection'* model, which was selected as a suitable starting point for real vs AI image classification [18]. This model was then trained on a combined dataset, built from NTIRE, JulienLucas and DeepfakeJudge image sources, split into training, validation and test sets using an 80/10/10 structure, respectively, while maintaining a 50/50 balance between real and AI-generated images [19][20][21]. The module returns a probabilistic visual AI score, which is also passed to the decision fusion stage. This output is valuable because visual analysis remains available even when metadata has been removed or is incomplete. However, it should not be interpreted as proof of synthetic origin. As discussed in the background, visual detectors may be affected by dataset shift, unseen generators, compression, cropping or post-processing, and their confidence scores only indicate similarity to patterns learned during training. Therefore, metadata and visual classification were treated as complementary rather than competing signals. Metadata can indicate provenance and software history, while the classifier assesses image content directly. Neither signal is sufficient on its own, so both are combined later through decision fusion to support a more transparent and cautious assessment.

3.3 Conditional Deepfake & Semantic Analysis

The conditional deepfake module is developed as a second layer of the interpretive stage, which is triggered only when the fused AI probability is higher than the selected threshold. It does not replace the metadata or visual classification modules, and it does not decide independently whether an image is AI-generated. Instead, it provides additional semantic context for images that have already been classified as AI-positive or suspicious. Once triggered, the module applies YuNet face detection to identify whether the image contains a human face and to estimate the confidence of the detected region [22][23]. Where a face is detected, the relevant face region is cropped and reclassified using the visual classifier. This allows the system to assess whether the facial area itself contributes to the AI-generated classification, rather than relying only on the full-image score. This is relevant because face-based content is closely connected to deepfake risks, particularly where synthetic media may imply the presence, likeness or identity of a real person.

The module also includes semantic and landmark-based analysis to support interpretation. DINOv2 embeddings are used to represent the image visually, while a FAISS landmark index allows the system to compare the image against stored landmark references and retrieve visually similar examples where available [24][25][26]. These outputs are additions to the final explanation, showing whether an AI-positive image contains face-like or place-like parts. However, they should not be interpreted as forensic proof of manipulation. The landmark retrieval relies on the index coverage. Scene classification could be ambiguous and face detection could be influenced by image quality or multiple faces. So the module helps with interpretation, rather than being a stand-alone deepfake verdict.

3.4 Decision Fusion, Explainability & Interface

Decision fusion was implemented using the reliability-weighted formula described below, to combine imperfect metadata and visual signals into a single interpretable assessment, preserving uncertainty. This was necessary because neither metadata nor visual classification is individually reliable enough to support a user-facing authenticity assessment.

The main fused AI probability was calculated using a reliability-weighted average:

$$P(AI) = \frac{(w_m\, a_m\, P(AI)_m) + (w_v a_v P(AI)_v^*)}{(w_m\, a_m) + (w_v a_v)}$$

where $P(AI)$ is the final fused probability that the image is AI-generated, $P(AI)_m$ is the probability derived from metadata and provenance indicators, and $P(AI)_v^*$ is the adjusted visual probability (explained below). The terms w_m and w_v represent the relative importance assigned to the metadata and visual modules, while a_m and a_v represent reliability or accuracy scaling factors. These allow weaker or unavailable signals to have less influence on the final score.

The adjusted visual probability was defined as:

$$P(AI)_v^* = \max\left(P(AI)_v, P(AI)_{vcf}\right)$$

where $P(AI)_v$ is the probability returned by the full-image visual classifier and $P(AI)_{vcf}$ is the probability returned when a detected face region is cropped and reclassified. This was included because a face region may contain stronger AI-generation artefacts than the wider image, especially where background content reduces the strength of the full-image prediction.

The fused probability was then compared against a threshold to produce the first stage classification:

$$\widehat{y_1} = \begin{cases} D_1, P(AI) < 0.5 \\ D_2, P(AI) \geq 0.5 \end{cases}$$

where D_1 denotes that the image is not AI-generated, and D_2 denotes that the image is AI-generated but not necessarily a deepfake.

Conditional on $\widehat{y_1} = D_2$, a second-stage rule was used to determine whether an AI-positive image should be described as a probable deepfake:

$$\widehat{y_2} = \begin{cases} D_2, (P_f < 0.75) \wedge (P_{lm} < 0.5) \\ D_3, (P_f \geq 0.75) \vee (P_{lm} \geq 0.5) \end{cases}$$

where D_3 denotes that the image is AI-generated and probably a deepfake, P_f represents the face-based confidence of YuNet, and P_{lm} represents the landmark confidence of DINOv2. This rule is consistent with the layered transparency approach of the Code of Practice, where marking, provenance, labelling and detection mechanisms are considered as complementary signals, rather than conclusive evidence.

The interface presents these outputs in a user-facing format. Rather than showing only a binary label, it exposes the metadata score, visual probability, fused confidence score, final verdict and explanatory information. Where available, heatmap-style explanations indicate which regions influenced the visual classifier, but these explanations should be treated as interpretive support rather than proof of manipulation. This reflects the literature review point that explainability is necessary for decision-support systems, since regulators, investigators and non-expert users need to understand how an assessment was reached rather than simply receive a final label. Overall, the system's final output is an evidence-based confidence assessment that exposes uncertainty and supports human interpretation.

4. Results & Evaluation

4.1 Evaluation Approach

The prototype was evaluated as a modular decision-support pipeline rather than as a single detector. This reflects the design of the system, where metadata analysis, visual classification, semantic analysis and decision fusion each contribute different forms of evidence. The evaluation thus considered the behaviour at the module level as well as the end-to-end output. The metadata module, visual classifier, and conditional semantic parts were tested at the module level to see if they generated valid results. End-to-end testing was then used to examine whether these outputs were combined effectively through the decision fusion mechanism.

4.2 Metadata & Visual Classification Results

The metadata module performed most strongly when explicit provenance or generator related traces were present. Images containing C2PA-style indicators, software fields or AI-related metadata were assigned high metadata-derived AI-origin scores. This behaviour was consistent with the purpose of the module, since such indicators provide more direct contextual evidence of possible AI generation than image appearance alone. Images with camera-style metadata, on the other hand, were given lower AI-origin scores; images with no or limited metadata generally received neutral scores. This was an important design choice, since missing metadata cannot reliably prove either authenticity or manipulation. Metadata may be removed during upload, compression or re-sharing, and therefore absence of evidence was treated as uncertainty rather than as evidence that an image was real. Annex II (1) presents the results of the complete metadata module.

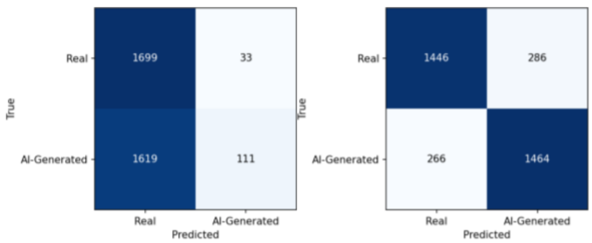

Figure 2: Confusion matrix comparison between the base model (Left) and the fine-tuned visual classifier (Right) on the combined test dataset.

The visual classifier provided the second main evidence stream by analysing the image content directly. The original base model performed poorly on the combined test dataset, showing a strong bias towards the real class (Figure 2). Although it classified many real images correctly, its AI-generated recall was very low, meaning that most AI-generated samples were missed. Fine-tuning substantially improved performance. These results suggest that the final model learned a more balanced decision boundary between real and AI-generated images. These results are shown in Annex II (2) and summarised in Figure 3.

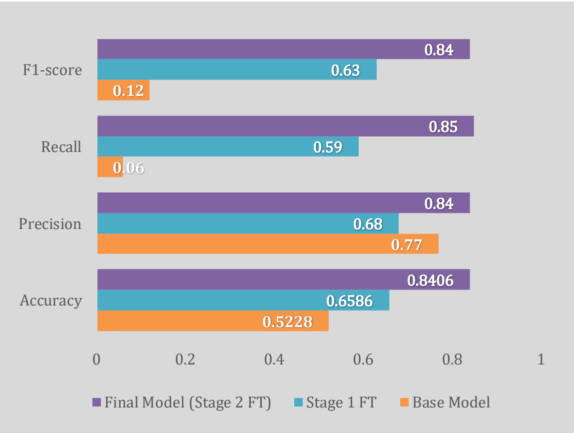

Figure 3: Performance comparison of the base model, Stage 1 fine-tuned model and final Stage 2 fine-tuned visual classifier on the combined test dataset.

However, the classifier result should still be treated cautiously. The evaluation was conducted on a controlled combined test split, not on open-world images collected from social media or future generative systems. Therefore, the model's improved performance demonstrates successful dataset adaptation, but not complete generalisation. This supports the wider design decision to treat the classifier as one probabilistic signal within a broader transparency pipeline, rather than as a standalone authenticity judgement.

4.3 Conditional Semantic Analysis

The conditional semantic module was evaluated on the controlled sample set containing different combinations of faces and places. The conditional deepfake analysis module was designed to operationalise part of the EU AI Act's broader definition of deepfake content. That definition contains content similar to existing persons, objects, places, entities or events. The prototype centred around two image-relevant dimensions, face-like and place-like content, that could be feasibly tested in the placement. This was not to classify an image as real or AI-generated but to add further interpretation once an image had already been treated as AI-positive or suspicious by the main pipeline. This distinction is important because the module was designed as an explanatory layer, not as an independent detector.

The face analysis component showed useful behaviour in several cases. Where clear face-like regions were present, YuNet returned high confidence scores, such as 0.9163 for the results of Figure 4. Where no face was present, YuNet returned very low confidence scores, such as 0.0001, and no face proposals were produced. This supported the conditional logic of the system, since face-based analysis should only be considered meaningful where a face-like region is actually detected. However, the results also showed a limitation as a stylised LEGO figure was detected as a face-like region. This suggests that the result should be taken as evidence for face-like visual structure rather than as evidence for the image containing or manipulating a real person.

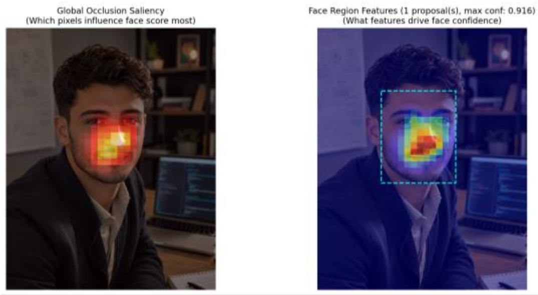

Figure 4: Conditional deepfake analysis output for an AI-generated face-like image.

The landmark component added further semantic context by using DINOv2 embeddings and a FAISS landmark index to identify place-like content. This was useful because synthetic media risks are not limited to faces, as per the legal definition, AI-generated images may also depict recognisable buildings, public places or event-like scenes [3]. Several place-containing samples produced landmark confidence values above the selected threshold, supporting their classification as possible deepfake-related content if the image had already been identified as AI-generated. Nevertheless, this module is also limited by the coverage of the landmark index and by the uncertainty of scene-level matching. The conditional results are therefore best read as interpretive support for human review rather than forensic proof of manipulation. The full results are provided in Annex II (3).

Figure 5: *Conditional deepfake analysis output for a landmark-containing image, showing place-related saliency.*

4.4 Decision Fusion Results

The final evaluation considered whether the fused pipeline produced coherent end-to-end outputs, such as the correct classifications across real, AI-generated and deepfake images. The decision fusion mechanism correctly classified all six real images as real and correctly classified the two non-face AI-generated images as AI-generated. This suggests that the combined metadata and visual signals were effective in avoiding false positives on the controlled sample set, while also detecting some AI-generated images that the original base model had previously struggled with.

Base Model

		Predicted	
		AI	Real
Actual	AI	4	2
	Real	4	2

Fused Pipeline

		Predicted	
		AI	Real
Actual	AI	2	4
	Real	0	6

Figure 6: *Comparison of base model and fused pipeline classifications on the controlled sample set.*

The main weakness was in the deepfake-style samples, where fusion consistently misclassified all four images as real. This failure was mainly caused by the fine-tuned visual classifier assigning very low AI probabilities to these images, despite some of them containing strong metadata evidence of AI origin. As a result, despite the high AI-origin score of the metadata module, the overall probability did not exceed the AI-generated threshold. This indicates that the selected fusion strategy was too strongly affected by the visual classifier when that classifier was confidently wrong. These results are shown in Annex II (4) and summarised in Figure 6.

This finding is central to the evaluation, as it highlights a significant limitation in the current fusion approach and suggests that the main thesis, that combining visual and metadata cues reliably improves AI image detection, may not hold in cases where the visual classifier is confidently incorrect. The results suggest that the visual classifier may not have generalised well to the more realistic deepfake-style samples. It has likely learned dataset specific features, possibly taking realistic faces or place-like content as evidence of real imagery. This does not prove the model just learned 'face equals real' but does show that face and place with synthetic images made a big failure case in the controlled sample set. Annex II (3) also shows that the semantic module identified several of these images as possible deepfakes if they were first accepted as AI-generated. Therefore, the semantic layer appeared to behave as intended, but the final system could not benefit from it because the earlier fusion stage suppressed the AI-positive decision.

5. Conclusion

5.1 Future Work

Future work should focus on improving the robustness of the visual classification component, as the model's failure to generalise reliably across deepfake-style samples was the most significant limitation identified during the evaluation. Although fine-tuning improved performance on the controlled dataset, the model did not generalise reliably across all deepfake-style samples, which means the evaluation results should be treated as indicative rather than conclusive. Further work should therefore examine alternative architectures to address the model's generalisation failures, more diverse training datasets to cover a wider range of deepfake styles, and more realistic evaluation conditions that better reflect real-world use. This should include image types that fell outside the current evaluation scope, such as images generated by newer models, compressed or re-saved images, edited images, screenshots, and images collected from platform-like environments.

Since this project will continue into the MSc dissertation, the framework could also be expanded beyond metadata analysis and visual classification. Further transparency signals such as watermark detection, perceptual hashing, and stronger provenance verification need to be investigated as they address forms of manipulation that may be missed by metadata and visual classification alone, given the existing literature and public tooling mature enough for proper evaluation. Future work should also extend the conditional semantic analysis beyond face-like and place-like content to include object recognition, entity-level identification and event-level interpretation, since these categories cover the range of manipulated content that deepfake-style analysis is most likely to encounter in practice. This would bring the framework closer to the broader definition of deepfake content in the EU AI Act, while still treating these signals as supporting, rather than conclusive, evidence of manipulation.

At present, some approaches, such as SynthID, are not publicly replicable in a way that would allow independent verification of their detection accuracy, as this would require open model weights or reproducible test data. Hence these methods should be seen as future signals for critical review, not as default reliable solutions.

5.2 Closing Remarks

This work placement showed that a proof-of-concept system can combine metadata analysis, visual classification, decision-level fusion and conditional semantic analysis to support AI-generated and deepfake image detection. The main contribution of the project is not a definitive solution to deepfake detection, but a practical framework for combining transparency signals, understanding their limitations, and supporting more informed judgements about suspicious content. Instead, the evaluation demonstrates that integrating multiple transparency signals can improve interpretability and highlight practical challenges, particularly where a confident but wrong classifier output overrode weaker metadata evidence.

The evaluation supports this claim, though the fusion limitation narrows where it applies rather than overturning the overall finding. Metadata was useful when available but could be missing or stripped, which left the fusion output relying solely on the visual classifier. The visual classifier worked better on controlled data but struggled with more realistic deepfake-style samples, showing the risk of poor generalisation. The fusion results also showed that combining signals can still be misleading when one evidence stream is confidently wrong.

Overall, given that fusion still misled when one signal was confidently wrong, the prototype should be understood as a regulatory learning and transparency-support tool rather than a final detector. It provides a platform for further MSc research into how imperfect evidence streams can be combined, calibrated and explained to support safer synthetic media assessment.

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Annex I: GitHub Repository & Implementation Evidence

The implementation developed during the work placement is available in the following GitHub repository:

https://github.com/DavidBScerri/Work-Placement-Assignment

The repository contains the prototype codebase used for the proof-of-concept system, including the visual classification module, metadata and provenance analysis module, conditional deepfake analysis components, decision-fusion pipeline and web-based interface. It also includes sample images, setup requirements and instructions for launching the prototype. The repository is provided as supporting evidence of the technical implementation described in the Methodology and Results sections of this report, rather than as a production-ready deployment package.

Repository accessed on: **30th June 2026 04:37am**
The submitted report reflects the repository state at the time of final submission.

Annex II: Quantitative Evaluation Results

(1) Metadata and provenance module results on controlled sample images

Sample Images	Metadata condition	$P(AI)_m$ score	Decision	Rationale / evidence detected	Outcome
ai+metadata -place-face.png	Present	0.99	likely_ai_generated	• Explicit AI-related generator or provenance signal found; • C2PA/content provenance-style metadata detected; • AI keyword hits: c2pa, claim_generator, openai; • binary marker hits: c2pa, claim_generator, openai; • software tags present without normal camera metadata.	Correct
ai-metadata -place-face.png	Absent/ Limited	0.50	uncertain	• No meaningful metadata or provenance rationale detected.	Correct
deepfake+metadata +place-face.png	Present	0.99	likely_ai_generated	• Explicit AI-related generator or provenance signal found; • C2PA/content provenance-style metadata detected; • AI keyword hits: c2pa, claim_generator, openai; • binary marker hits: c2pa, claim_generator, openai; • software tags present without normal camera metadata.	Correct
deepfake+metadata -place+face.png	Present	0.99	likely_ai_generated	• Explicit AI-related generator or provenance signal found; • C2PA/content provenance-style metadata detected; • AI keyword hits: c2pa, claim_generator, openai; • binary marker hits: c2pa, claim_generator, openai; • software tags present without normal camera metadata.	Correct
deepfake-metadata +place-face.png	Absent/ Limited	0.53	uncertain	• Limited binary marker evidence detected: c2pa.	Correct
deepfake-metadata -place+face.png	Absent/ Limited	0.50	uncertain	• No meaningful metadata or provenance rationale detected.	Correct
real+metadata +place-face.jpg	Present	0.35	likely_camera_origin	• Timestamp pattern flagged as unusually clean/perfect; • rich camera metadata present.	Correct
real+metadata -place+face.jpeg	Present	0.35	likely_camera_origin	• Timestamp pattern flagged as unusually clean/perfect; • rich camera metadata present.	Correct
real+metadata -place-face.jpeg	Present	0.35	likely_camera_origin	• Timestamp pattern flagged as unusually clean/perfect; • rich camera metadata present.	Correct
real-metadata +place-face.jpg	Absent/ Limited	0.50	uncertain	• No meaningful metadata or provenance rationale detected.	Correct
real-metadata -place+face.jpg	Absent/ Limited	0.50	uncertain	• No meaningful metadata or provenance rationale detected.	Correct
real-metadata -place-face.jpg	Absent/ Limited	0.50	uncertain	• No meaningful metadata or provenance rationale detected.	Correct

(2) Visual classifier performance on the combined test dataset

Model stage	Accuracy	Precision	Recall	F1-score
Original base model	52.28%	0.77	0.06	0.12
Stage 1 fine-tuning	65.86%	0.68	0.59	0.63
Stage 2 fine-tuning	84.06%	0.84	0.85	0.84

(3) Conditional deepfake analysis results on sample images

Sample Images	Face confidence (P_f)	Landmark confidence (P_{lm})	Final classification	Outcome
ai+metadata -place-face.png	0.0001	0.1905	Not a Possible Deepfake	Correct
ai-metadata -place-face.png	0.0001	0.1905	Not a Possible Deepfake	Correct
deepfake+metadata +place-face.png	0.0001	0.5501	Possible Deepfake (if AI Generated)	Correct
deepfake+metadata -place+face.png	0.9163	0.2909	Possible Deepfake (if AI Generated)	Correct
deepfake-metadata +place-face.png	0.0001	0.5501	Possible Deepfake (if AI Generated)	Correct
deepfake-metadata -place+face.png	0.9163	0.2909	Possible Deepfake (if AI Generated)	Correct
real+metadata +place-face.jpg	0.0001	0.6214	Possible Deepfake (if AI Generated)	N/A
real+metadata -place+face.jpeg	0.9372	0.4051	Possible Deepfake (if AI Generated)	N/A
real+metadata -place-face.jpeg	0.5005	0.2309	Not a Possible Deepfake	N/A
real-metadata +place-face.jpg	0.0001	0.6195	Possible Deepfake (if AI Generated)	N/A
real-metadata -place+face.jpg	0.9374	0.4086	Possible Deepfake (if AI Generated)	N/A
real-metadata -place-face.jpg	0.6814	0.1625	Not a Possible Deepfake	N/A

(4) Decision-fusion results on controlled sample images

Sample Images	Expected	$P(AI)_m$ score	$P(AI)_v$ score	Fused $P(AI)$ score	Fused label	Outcome
ai+metadata -place-face.png	AI Generated	0.99	0.8264	0.8695	AI Generated	Correct
ai-metadata -place-face.png	AI Generated	0.50	0.8264	0.7405	AI Generated	Correct
deepfake+metadata +place-face.png	Deepfake	0.99	0.0625	0.3066	Real	Incorrect
deepfake+metadata -place+face.png	Deepfake	0.99	0.0668	0.3097	Real	Incorrect
deepfake-metadata +place-face.png	Deepfake	0.53	0.0625	0.1855	Real	Incorrect
deepfake-metadata -place+face.png	Deepfake	0.50	0.0668	0.1808	Real	Incorrect
real+metadata +place-face.jpg	Real	0.35	0.1460	0.1997	Real	Correct
real+metadata -place+face.jpeg	Real	0.35	0.2917	0.3070	Real	Correct
real+metadata -place-face.jpeg	Real	0.35	0.1147	0.1766	Real	Correct
real-metadata +place-face.jpg	Real	0.50	0.2168	0.2913	Real	Correct
real-metadata -place+face.jpg	Real	0.50	0.3876	0.4172	Real	Correct
real-metadata -place-face.jpg	Real	0.50	0.1978	0.2773	Real	Correct

Annex III: Qualitative Visual Analysis Results

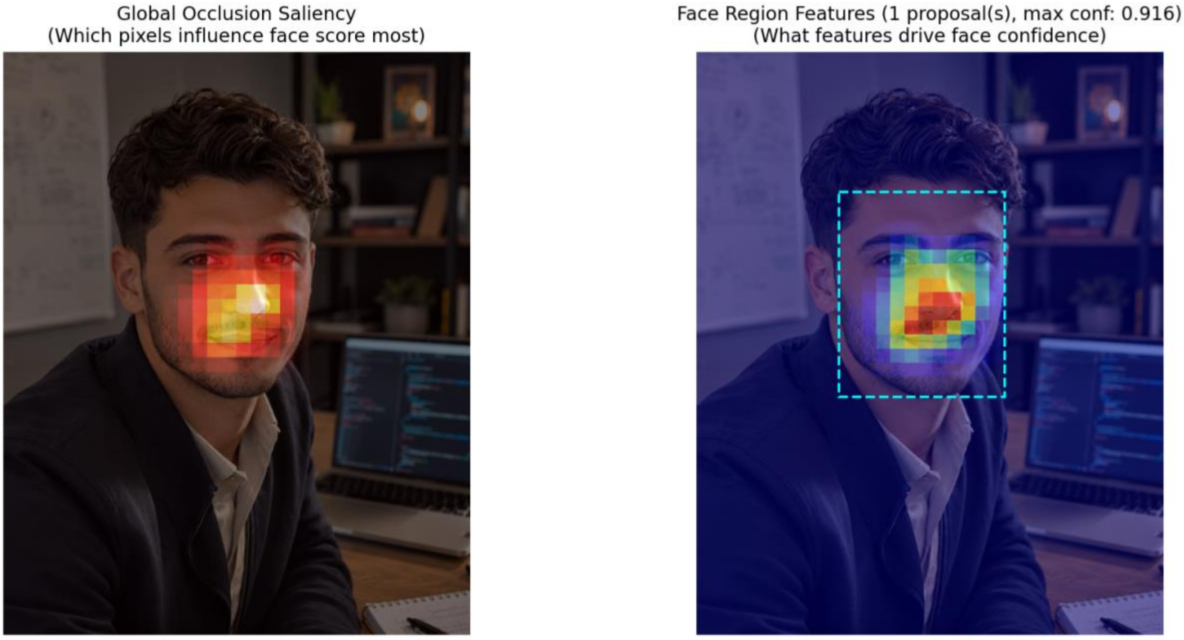
(1) Face Analysis Visualisations

<u>Figure 1.1:</u> Face analysis for a non-face AI-generated image.

4. Very low confidence (0.0001);
5. no meaningful face detected.

<u>Figure 1.2:</u> Face saliency for a portrait-style deepfake sample.

6. High confidence (0.9163);
7. saliency is concentrated on the central facial region, indicating a strong face detection;
8. same image as Figure 4 in section 4.3.

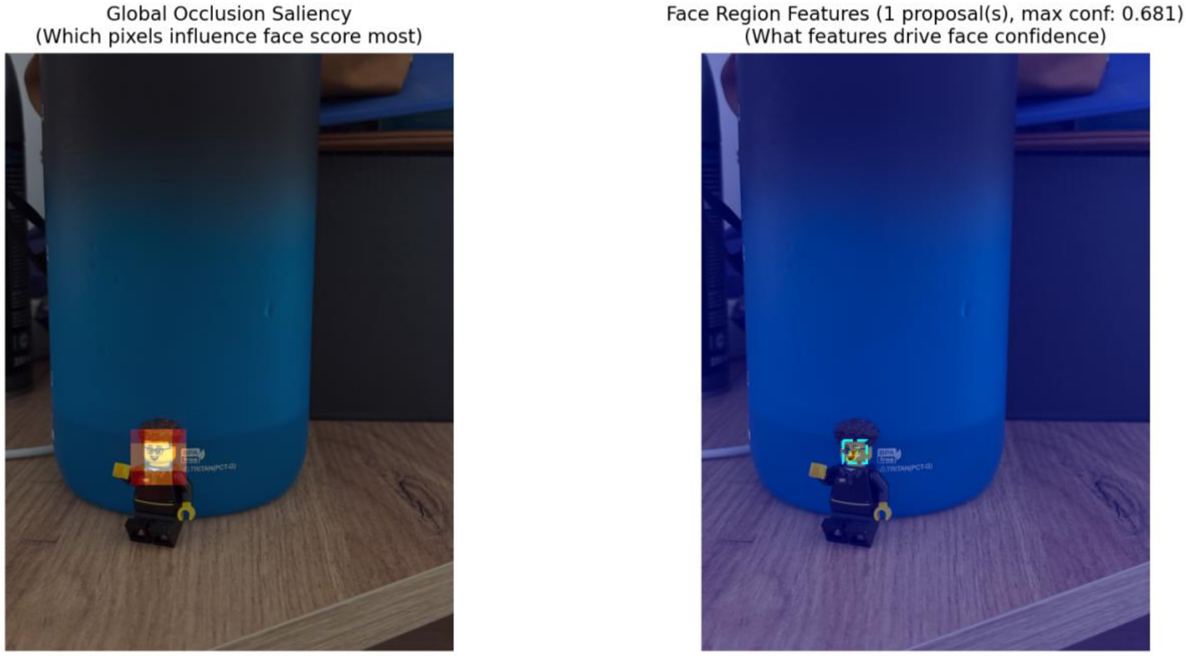


<u>Figure 1.3:</u> Face analysis for an AI-generated place image.

9. Very low confidence (0.0001);
10. correctly produces no face proposals.

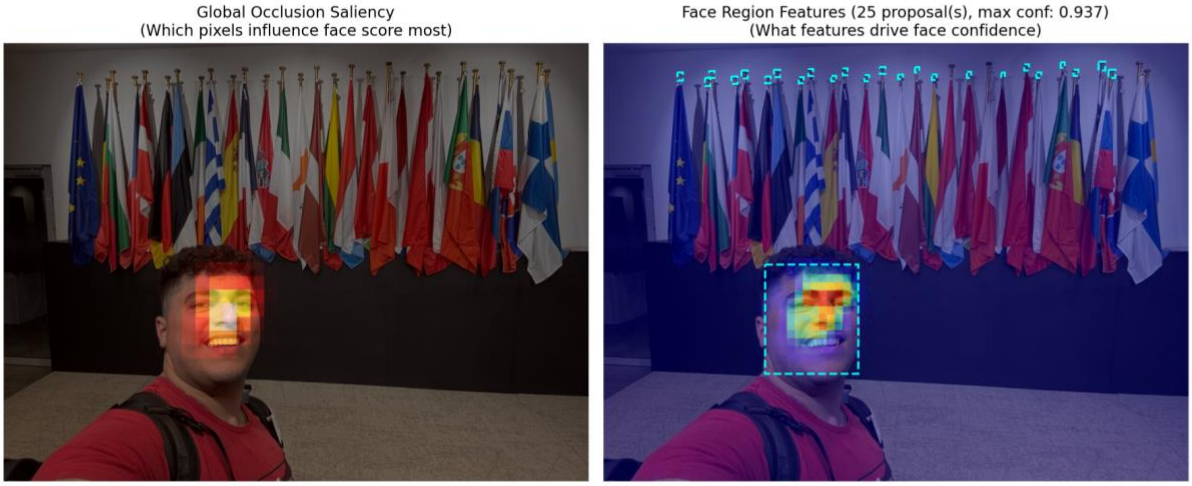
<u>Figure 1.4:</u> Face analysis for a real image containing a stylised LEGO figure.

11. Moderate confidence (0.6814);
12. the module detects the LEGO face as a face-like structure, illustrating a false positive.



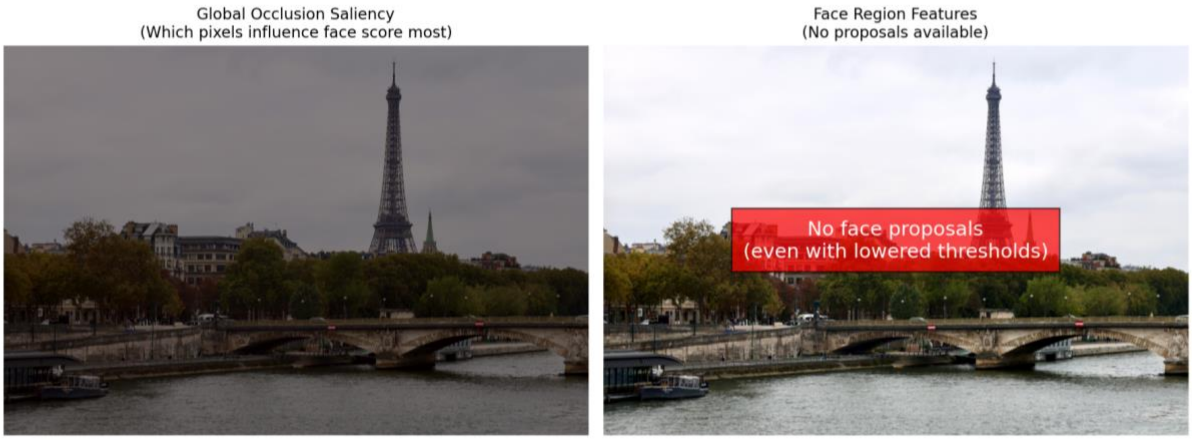
<u>Figure 1.5:</u> Face saliency for a real image containing a human subject.

13. High confidence (0.9374);
14. saliency and proposal regions align with the subject's face, indicating a strong face detection.



<u>Figure 1.6:</u> Face analysis for a real place image without a human subject.

15. Very low confidence (0.0001);
16. correctly produces no face proposals.



(2) Landmark Analysis Visualisations

***Figure 2.1:* Landmark analysis for a non-place AI-generated image.**

- Low confidence (0.1905);
- correctly classified as unknown.

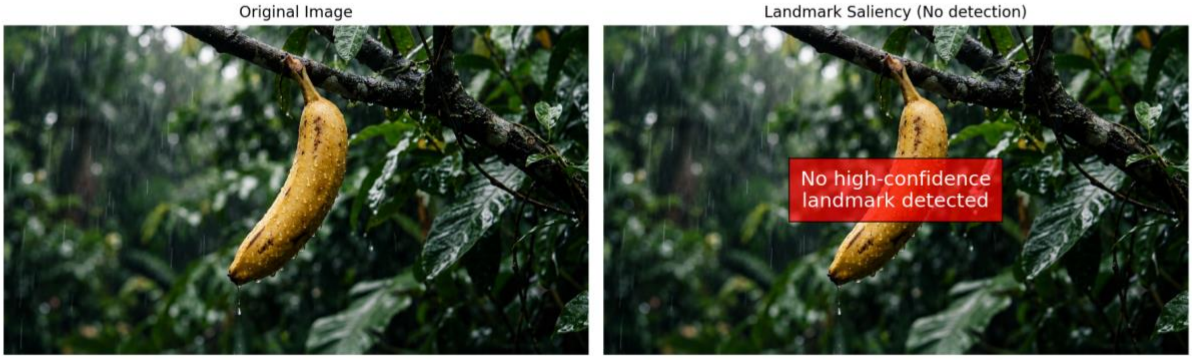


***Figure 2.2:* Landmark analysis for a portrait-style deepfake sample.**

- Low confidence (0.2909);
- no landmark detected.

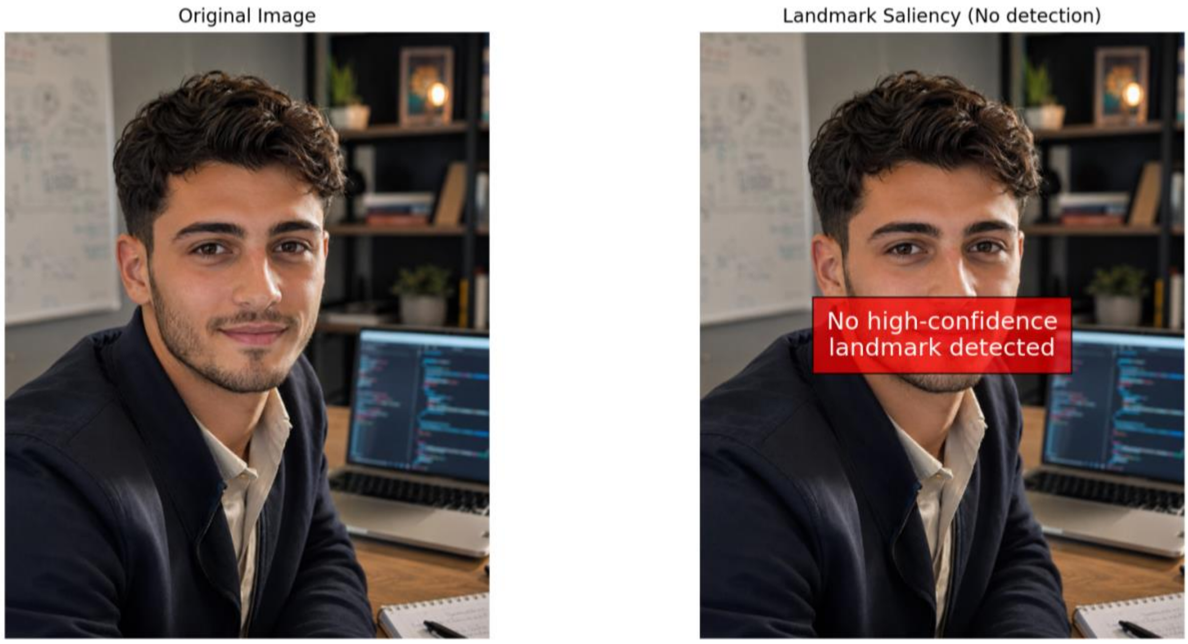


***Figure 2.3:* Landmark saliency for an AI-generated place image.**

- Moderate confidence (0.5501);
- saliency focuses on scene structure.

***Figure 2.4:* Landmark analysis for a real image containing a stylised LEGO figure.**

17. Low confidence (0.1625);
18. correctly classified as unknown.

<u>Figure 2.5:</u> Landmark analysis for a real image containing a person and background flags.

19. Moderate-low confidence (0.4086);
20. classified as unknown.

<u>Figure 2.6:</u> Landmark saliency for a real place image.

21. Higher confidence (0.6195);
22. saliency highlights key landmark features;
23. same image as Figure 5 in section 4.3.